

Tarea Sección 5.1

3) a) 620°

Co-terminales positivos

$$\begin{aligned} 620^\circ + 360^\circ &= 980^\circ \\ 980^\circ + 360^\circ &= 1340^\circ \end{aligned}$$

Co-terminales negativos

$$\begin{aligned} 620^\circ - 360^\circ &= 260^\circ \\ 260^\circ - 360^\circ &= -100^\circ \end{aligned}$$

b) $\frac{5\pi}{6}$

Positivos

$$\frac{5\pi}{6} + 2\pi = \frac{17\pi}{6}$$

$$\frac{17\pi}{6} + 2\pi = \frac{29\pi}{6}$$

Negativos

$$\frac{5\pi}{6} - 2\pi = -\frac{7\pi}{6}$$

$$-\frac{7\pi}{6} - 2\pi = -\frac{19\pi}{6}$$

c) $-\frac{\pi}{4}$

Positivo

$$-\frac{\pi}{4} + 2\pi = \frac{7\pi}{4}$$

$$\frac{7\pi}{4} + 2\pi = \frac{15\pi}{4}$$

Negativo

$$-\frac{\pi}{4} - 2\pi = -\frac{9\pi}{4}$$

$$-\frac{9\pi}{4} - 2\pi = -\frac{17\pi}{4}$$

$$5) a) \theta = 120^\circ 37' 24'' = \underline{77^\circ 22' 36''}$$

$$b) \theta = 43.87^\circ \quad 90^\circ - 43.87^\circ = \underline{46.13^\circ}$$

$$7) a) \theta = 125^\circ 16' 27'' = \underline{51^\circ 43' 33''}$$

$$b) \theta = 58.07^\circ \quad 180^\circ - 58.07^\circ = \underline{121.93^\circ}$$

$$11) a) 450^\circ$$

$$450^\circ \cdot \frac{\pi}{180^\circ} = \underline{\frac{5\pi}{2}}$$

$$b) 72^\circ$$

$$72^\circ \cdot \frac{\pi}{180^\circ} = \underline{\frac{2\pi}{5}}$$

$$c) 100^\circ$$

$$100^\circ \cdot \frac{\pi}{180^\circ} = \underline{\frac{5\pi}{9}}$$

$$15) a) -\frac{7\pi}{2}$$

$$-\frac{7\pi}{2} \cdot \frac{180^\circ}{\pi} = \underline{-630^\circ}$$

$$b) 7\pi$$

$$7\pi \cdot \frac{180^\circ}{\pi} = \underline{1260^\circ}$$

$$c) \frac{\pi}{9}$$

$$\frac{\pi}{9} \cdot \frac{180}{\pi} = \underline{20^\circ}$$

$$21) \theta = 3.7$$

$$3 \left(\frac{180^\circ}{\pi} \right) = 177.6769$$

$$177^\circ + (0.6769)(60') = 37.014$$

$$177^\circ + 37' + (0.014)(60'') = 0.84$$

$$\underline{177^\circ 37' 0.84''}$$

$$23) 262^\circ 15' 31''$$

$$262^\circ + \left(\frac{15}{60} \right)^\circ + \left(\frac{31}{3,600} \right)$$

$$262^\circ + 0.25^\circ + 0.0086 = \underline{262.2586}$$

$$27) 370.6275^\circ = \underline{370^\circ 37' 17.4''}$$

$$0.6275 \cdot 60 = 37.29$$

$$0.29 \cdot 60 = 17.4$$

$$29) s = 10\text{cm} \quad \theta = 4$$

$$10\text{cm} = r \cdot 4$$

$$\frac{10}{4} = r$$

$$4$$

$$r = \underline{2.5\text{cm}}$$

$$31) r = 8\text{cm}$$

$$\theta = 45^\circ$$

$$45^\circ \neq \frac{\pi}{180^\circ} = \frac{\pi}{4}$$

$$a) S = (8\text{cm}) \left(\frac{\pi}{4} \right)$$

$$S = \underline{2\pi}$$

$$b) A = \frac{1}{2} (8\text{cm})^2 \left(\frac{\pi}{4} \right)$$

$$A = \frac{1}{2} (64\text{cm}^2) \left(\frac{\pi}{4} \right)$$

$$A = 32\text{cm}^2 \left(\frac{\pi}{4} \right)$$

$$A = \underline{25.13}$$

$$33) S = 7\text{cm}$$

$$r = 4\text{cm}$$

$$a) 7\text{cm} = 4\text{cm} \theta$$

$$7/4 = \theta = \underline{1.75}$$

$$1.75 \neq \frac{180^\circ}{\pi} \approx \underline{100.27^\circ}$$

$$b) A = \frac{1}{2} (4\text{cm})^2 (1.75)$$

$$A = \frac{1}{2} (16\text{cm}^2) (1.75)$$

$$A = 8 (1.75)$$

$$A = \underline{14\text{cm}^2}$$

$$35) \theta = 50^\circ$$

$$d = 16\text{m}$$

$$r = 8\text{m}$$

$$\theta = 50^\circ \neq \frac{\pi}{180^\circ} = \frac{5\pi}{18}$$

$$S = 8\text{cm} \left(\frac{5\pi}{18} \right)$$

$$S = \underline{\frac{20\pi}{9}} \approx \underline{6.98\text{m}}$$

$$A = \frac{1}{2} (8\text{m})^2 \left(\frac{5\pi}{18} \right)$$

$$A = 32\text{m}^2 \left(\frac{5\pi}{18} \right)$$

$$A = \underline{27.93}$$

$$37) S = 8$$

$$S = r\theta$$

$$r = \frac{S}{\theta}$$

$$A = \frac{1}{2} r^2 \theta$$

$$A = \frac{1}{2} \left(\frac{8}{\theta} \right)^2 \theta$$

$$A = \frac{1}{2} \theta \cdot \frac{64}{\theta^2} \cdot \theta$$

$$A = 32/\theta$$

43) Área ventana

$$A = l \cdot a$$

$$A = 54 \cdot 24$$

$$A = 1,296 \text{ in} = 100\%$$

③ Área no limpiada

$$A = \pi (5)^2 (120)$$

$$A = 26.81 \text{ in}^2$$

$$\begin{array}{rcl} ⑤ & 1,296 & \text{---} 100\% \\ & 480.66 & \text{---} X \end{array}$$

$$X = \frac{100 \cdot 480.66}{1,296} = 37.1\%$$

$$45) 23^\circ 56' 4'' \approx 23.93^\circ \cdot \frac{\pi}{180} = 0.42 \text{ rad/s}$$

② Área limpiador

$$A = \frac{\pi r^2 \theta}{360}$$

$$r = 17.5$$

$$r = 22$$

$$A = \frac{\pi (22)^2 (120)}{360}$$

$$A = 506.84 \text{ in}^2$$

④ Área total limpiada

$$506.84 - 26.78 = 480.06 \text{ in}^2$$

49)

$$a) r \omega = v$$

$$\omega (200 \text{ r/m}) \left(\frac{2\pi \text{ rad}}{1 \text{ r}} \right) = \underline{400\pi \text{ rad/min}}$$

$$b) 200 \text{ r/m} \cdot \frac{1 \text{ m}}{60 \text{ s}} = 3.33 \text{ r/s} \cdot \frac{2\pi \text{ rad}}{1 \text{ r}} = 6.66\pi \text{ rad/s}$$

$$5.7 \text{ cm} \cdot 6.66\pi \text{ rad/s} = \underline{38 \text{ años}}$$

$$c) \omega = \frac{v}{r} = \frac{38}{3} = 12.67 \text{ rad/s} \cdot \frac{1 \text{ rad}}{2\pi \text{ rad}} \cdot \frac{60}{1} = \underline{380 \text{ rpm}}$$

$$d) \omega(r) = \frac{1.140}{r} = \text{inversa}$$

$$53) \frac{27}{4}$$

$$\frac{169}{16}$$

h grande

pequeña = p

grande = g

$$p = 3,376.75$$

$$g = 10.563.52$$

$$55) r_1 = 5 \text{ in} \quad r_2 = 2 \text{ in}$$

$$d = 28 \text{ in} \quad \text{rpm} = ?$$

$$V = 40 \text{ mi/h} = 704 \text{ in/s}$$

$$(5 \text{ in}) \omega = 704 \text{ in/s}$$

$$\omega = 140.8 \text{ r/s}$$

$$\frac{140.8 \text{ r}}{1 \text{ s}} \cdot \frac{60 \text{ seg}}{1 \text{ min}} = \underline{8,448 \text{ rpm}}$$